

Joint high-dimensional Bayesian variable and covariance selection with an application to eQTL analysis

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- Variable and (inverse) covariance selections have been well-studied separately in high-dimensional problems.
- However, “joint” selection (or estimation) have not been studied until recently.
- We formulate a Bayesian technique and apply it to the analysis of expression quantitative trait loci (eQTL) analysis.
- Joint work with Bani K. Mallick, Texas A&M University.

Problem Formulation

- n = Sample size.
- X = An $n \times p$ matrix of predictors.
- Y = An $n \times q$ matrix of responses.
- We would like to regress Y on X .
- Example A: For the same n individuals, we might try to see how their **SNP genotype** (X) affect their **gene expressions** (Y).
- Example B: For the same n individuals with cancer, we might try to see how their **microRNA** (X) affect their **mRNA** (Y) expressions.
- I have worked on A; I plan to begin work on B.

Problem Formulation

- Consider the linear Gaussian regression model:

$$\begin{aligned}\mathbf{Y}_{n \times q} &= \mathbf{X}_{n \times p} \mathbf{B}_{p \times q} + \boldsymbol{\epsilon}_{n \times q}, \\ \boldsymbol{\epsilon}_{n \times q} &\sim \text{MN}_{n \times q}(\mathbf{0}, \mathbf{I}_n, \boldsymbol{\Sigma}_{q \times q}), \\ \text{i.e., } \text{Vec}(\boldsymbol{\epsilon}_{n \times q}) &\sim \text{N}_{nq}(\mathbf{0}, \mathbf{I}_n \otimes \boldsymbol{\Sigma}_{q \times q}).\end{aligned}$$

- The unknowns are $\mathbf{B}_{p \times q}$ and $\boldsymbol{\Sigma}_{q \times q}$.
- The dimensions are pq and $q(q+1)/2$. Often much larger than n .
- Typical values: $n = 100$, $p = 500$ to 3000 , $q = 100$.

Basics of variable and covariance selection

- When p and q are larger than n , it becomes necessary to determine a **sparse** set of predictors and inverse covariance matrix elements.
- Variable selection: Find out the important predictors.
 - Typical assumption: Errors are i.i.d (i.e., $\Sigma_{q \times q} = \sigma^2 \mathbf{I}_q$).
- Covariance selection: Find out the important inverse covariance matrix elements.
 - For Gaussian models: $\Sigma_{i,j}^{-1} = 0 \iff Y_i \perp Y_j | \text{rest}$.
 - Typical assumption: No covariates (i.e., $\mathbf{B}_{p \times q} = 0$).
- We do a joint selection. This is being done only recently.

Previous Work in variable selection

- Variable selection with i.i.d errors.
- Frequentist: Lasso (Tibshirani, 1996, JRSSB) and its various extensions using ℓ_1 penalty.
- Bayesian: Stochastic Search Variable Selection (George and McCulloch, 1997, JASA) and its extensions using sparsity prior.

Previous Work in covariance selection and estimation

- (Inverse) Covariance selection in Gaussian graphical model with zero mean.
- Frequentist: Meinshausen and Bühlmann (2006, Ann. Stat.), Graphical Lasso (Friedman et al, 2008, Biostatistics), Bickel and Levina (2008, Ann. Stat.) etc.
- Bayesian: Carvalho and West (2007, Biometrika) etc. primarily using hyper-inverse Wishart type of priors.

Joint modeling of mean and covariance for Seemingly Unrelated Regression

- In a Seemingly Unrelated Regression setting, one might be interested in modeling “both” the mean and the covariance structure.
- Rothman et al. (2010, JCGS) make a frequentist attempt at joint modeling with the MRCE approach. (essentially an iterative approach with alternating `lasso()` and `glasso()` steps).
- Yin and Li (2011, Ann. Appl. Stat.) apply a similar approach to gene expression and SNP data.
- Bhadra and Mallick (Biometrics, under revision) take a Bayesian approach.

Model conditional on indicators

- Consider the model conditional upon indicators γ and $\mathbf{G}$.

$$\mathbf{Y} = \mathbf{X}_\gamma \mathbf{B}_{\gamma, \mathbf{G}} + \epsilon, \quad \epsilon \sim \text{MN}(\mathbf{0}, \mathbf{I}_n, \mathbf{\Sigma}_{\mathbf{G}})$$

- Dimension of $\mathbf{X}_\gamma = n \times p_\gamma$; dimension of $\mathbf{B}_{\gamma, \mathbf{G}} = p_\gamma \times q$;
dimension of $\mathbf{\Sigma}_{\mathbf{G}} = q \times q$.
- $\gamma_i = 1 \Rightarrow \mathbf{B}_{i, \cdot} \neq \mathbf{0}$; $p_\gamma = \sum_{i=1}^p \gamma_i$.
- $\mathbf{G}$ is a decomposable graph where $\mathbf{G}_{i,j} = 1 \Rightarrow \mathbf{\Sigma}_{i,j}^{-1} \neq 0$ with $i \neq j$; $i, j = 1, \dots, q$.

Model conditional on indicators: Toy example

- Consider the model conditional upon indicators γ and $\mathbf{G}$.

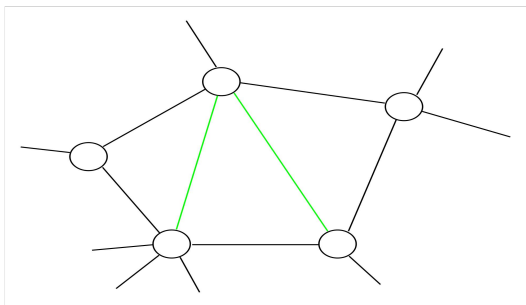
$$\mathbf{Y} = \mathbf{X}_\gamma \mathbf{B}_{\gamma, \mathbf{G}} + \epsilon; \quad \epsilon \sim \text{MN}(\mathbf{0}, \mathbf{I}_n, \boldsymbol{\Sigma}_{\mathbf{G}}).$$

- For example, say $p = q = 4$. Then $\gamma = (1, 0, 1, 0)$ means only the first and the third predictors are important.
- Let's say $\mathbf{G}$ is:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This means $\boldsymbol{\Sigma}_{1,2}^{-1} \neq 0$, the other off-diagonal terms are 0.

Decomposable (or triangulated) graphs



- No chordless cycle of length ≥ 3 .
- Cliques (i.e., the connected components) and separators (i.e., the parts in common between two cliques) can be found in polynomial time (NP-complete for general graphs).
- The overall density splits as:
$$f(y) = \prod_{j=1}^k f(y_{C_j}) / \prod_{j=2}^k f(y_{S_j}).$$

Bayesian hierarchical model

$$\begin{aligned}(\mathbf{Y} - \mathbf{X}_\gamma \mathbf{B}_{\gamma, \mathbf{G}}) | \mathbf{B}_{\gamma, \mathbf{G}}, \boldsymbol{\Sigma}_{\mathbf{G}} &\sim \text{MN}_{n \times q}(\mathbf{0}, \mathbf{I}_n, \boldsymbol{\Sigma}_{\mathbf{G}}), \\ \mathbf{B}_{\gamma, \mathbf{G}} | \gamma, \boldsymbol{\Sigma}_{\mathbf{G}} &\sim \text{MN}_{p_\gamma \times q}(\mathbf{0}, c \mathbf{I}_{p_\gamma}, \boldsymbol{\Sigma}_{\mathbf{G}}), \\ \boldsymbol{\Sigma}_{\mathbf{G}} | \mathbf{G} &\sim \text{HIW}_{\mathbf{G}}(b, d \mathbf{I}_q), \\ \gamma_i &\stackrel{\text{i.i.d}}{\sim} \text{Ber}(w_\gamma) \text{ for } i = 1, \dots, p, \\ G_k &\stackrel{\text{i.i.d}}{\sim} \text{Ber}(w_G) \text{ for } k = 1, \dots, q(q-1)/2, \\ w_\gamma, w_G &\sim \text{Uniform}(0, 1).\end{aligned}$$

Mariginalization of $B_{\gamma, \mathbf{G}}$ and $\Sigma_{\mathbf{G}}$

- Remember from the last slide

$$\begin{aligned}\epsilon &\sim \text{MN}_{n \times q}(\mathbf{0}, \mathbf{I}_n, \Sigma_{\mathbf{G}}), \\ \mathbf{B}_{\gamma, \mathbf{G}} | \gamma, \Sigma_{\mathbf{G}} &\sim \text{MN}_{p_{\gamma} \times q}(\mathbf{0}, c \mathbf{I}_{p_{\gamma}}, \Sigma_{\mathbf{G}}). \\ \Rightarrow \mathbf{X}_{\gamma} \mathbf{B}_{\gamma, \mathbf{G}} | \gamma, \Sigma_{\mathbf{G}} &\sim \text{MN}_{n \times q}(0, c(\mathbf{X}_{\gamma} \mathbf{X}_{\gamma}'), \Sigma_{\mathbf{G}}). \\ \Rightarrow \mathbf{Y} | \gamma, \Sigma_{\mathbf{G}} &\sim \text{MN}_{n \times q}(0, \mathbf{I}_n + c(\mathbf{X}_{\gamma} \mathbf{X}_{\gamma}'), \Sigma_{\mathbf{G}}).\end{aligned}$$

- Define $\mathbf{T} = \mathbf{A}\mathbf{Y}$ where $\mathbf{A}\mathbf{A}' = (\mathbf{I}_n + c(\mathbf{X}_{\gamma} \mathbf{X}_{\gamma}'))^{-1}$.

$$\begin{aligned}\Rightarrow \mathbf{T} | \gamma, \Sigma_{\mathbf{G}} &\sim \text{MN}_{n \times q}(\mathbf{0}, \mathbf{I}_n, \Sigma_{\mathbf{G}}). \\ \Sigma_{\mathbf{G}} | \mathbf{G} &\sim \text{HIW}_{\mathbf{G}}(b, d \mathbf{I}_q). \\ \Rightarrow \mathbf{T} | \gamma, \mathbf{G} &\sim \text{HMT}_{\mathbf{G}}(b, \mathbf{I}_n, d \mathbf{I}_q).\end{aligned}$$

The marginalized model

- After the marginalization of $\mathbf{B}_{\gamma, \mathbf{G}}$ and $\mathbf{\Sigma}_{\mathbf{G}}$, the resultant distribution is a “hyper matrix t”.
- This is a special type of “t-distribution” whose density splits over cliques and separators, given the graph.
- The marginalization has now resulted in a collapsed Gibbs sampler: need to sample only two quantities (γ and $\mathbf{G}$) instead of four ($\mathbf{B}_{\gamma, \mathbf{G}}$, $\mathbf{\Sigma}_{\mathbf{G}}$, γ and $\mathbf{G}$).
- Terms that were integrated out can always be sampled at the posterior, since we are working in a conjugate framework.

MCMC for γ given $\mathbf{G}$ and $\mathbf{T}$

- 1 Given the current γ , propose γ^* by either (a) changing a non-zero entry in γ to zero with probability $(1 - \alpha_\gamma)$ or (b) changing a zero entry in γ to one, with probability α_γ .
- 2 Calculate $f(\mathbf{t}|\gamma^*, \mathbf{G})$ and $f(\mathbf{t}|\gamma, \mathbf{G})$ where f denotes the HMT density.
- 3 Jump from γ to γ^* with probability

$$r(\gamma, \gamma^*) = \min \left\{ 1, \frac{f(\mathbf{t}|\gamma^*, \mathbf{G})p(\gamma^*)q(\gamma|\gamma^*)}{f(\mathbf{t}|\gamma, \mathbf{G})p(\gamma)q(\gamma^*|\gamma)} \right\}.$$

MCMC for $\mathbf{G}$ given γ and $\mathbf{T}$

- 1 Given the current $\mathbf{G}$, propose $\mathbf{G}^*$ by either (a) changing a non-zero edge in $\mathbf{G}$ to zero with probability $(1 - \alpha_G)$ or (b) changing a zero entry in $\mathbf{G}$ to one, with probability α_G .
- 2 Calculate $f(\mathbf{t}|\gamma, \mathbf{G}^*)$ and $f(\mathbf{t}|\gamma, \mathbf{G})$ where f denotes the HMT density.
- 3 Jump from $\mathbf{G}$ to $\mathbf{G}^*$ with probability

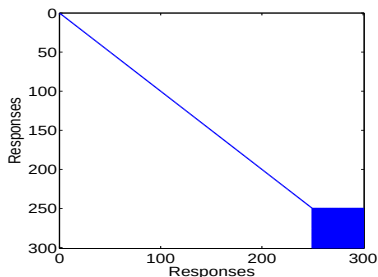
$$r(\mathbf{G}, \mathbf{G}^*) = \min \left\{ 1, \frac{f(\mathbf{t}|\mathbf{G}^*, \gamma)p(\mathbf{G}^*)q(\mathbf{G}|\mathbf{G}^*)}{f(\mathbf{t}|\mathbf{G}, \gamma)p(\mathbf{G})q(\mathbf{G}^*|\mathbf{G})} \right\}.$$

Regeneration of $\mathbf{B}_{\gamma, \mathbf{G}}$ in the posterior

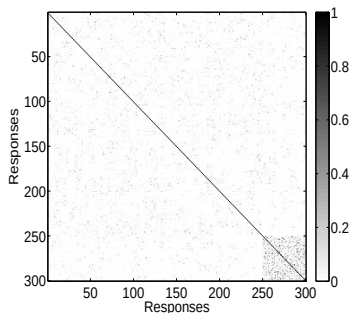
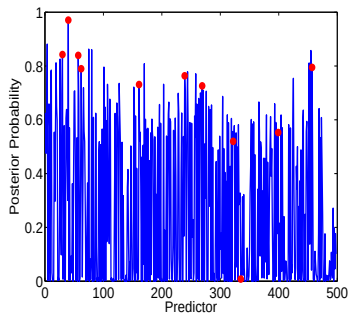
- $\mathbf{B}_{\gamma, \mathbf{G}}$ is the $p_\gamma \times q$ matrix of regression coefficients.
- By marginalizing it out we lose the association between the SNPs and expression levels necessary for an eQTL analysis.
- However, due to the conjugate structure, can be regenerated in the posterior conditional on $\hat{\gamma}$ and $\hat{\mathbf{G}}$.
- Generate $\Sigma_G | \mathbf{Y}, \mathbf{B}_{\gamma, \mathbf{G}}, \gamma, G$ from $\text{HIW}_G\{b + n, d\mathbf{I}_q + (\mathbf{Y} - \mathbf{X}_\gamma \mathbf{B}_{\gamma, \mathbf{G}})'(\mathbf{Y} - \mathbf{X}_\gamma \mathbf{B}_{\gamma, \mathbf{G}})\}$.
- Generate $\mathbf{B}_{\gamma, \mathbf{G}} | \mathbf{Y}, \Sigma_G, \gamma, G$ from $\text{MN}_{p_\gamma \times q}\{(\mathbf{X}'_\gamma \mathbf{X}_\gamma + c^{-1}\mathbf{I}_{p_\gamma})^{-1}\mathbf{X}'_\gamma \mathbf{Y}, (\mathbf{X}'_\gamma \mathbf{X}_\gamma + c^{-1}\mathbf{I}_{p_\gamma})^{-1}, \Sigma_G\}$.

Simulation study 1

- We choose $p = 498$, $q = 300$ and $n = 120$.
- The eleven true predictors are $\{30, 40, 57, 62, 161, 239, 269, 322, 335, 399, 457\}$.
- True adjacency matrix for **G** is shown below.

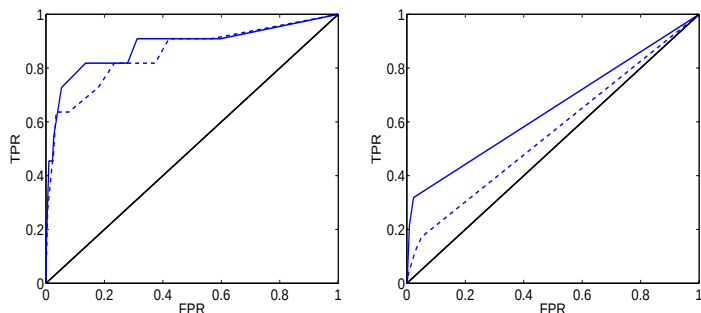


Results: Posterior probabilities



- Left: Posterior probabilities for γ , true variables circled in red.
- Right: Posterior probabilities for $\mathbf{G}$, compare with true graph.

Results: Does joint selection help over individual selection of variables and covariances?



- Left: ROC curve for γ , solid line: joint estimation, broken line: diagonal graph.
- Right: ROC curve for $\mathbf{G}$, solid line: joint estimation, broken line: zero mean model.

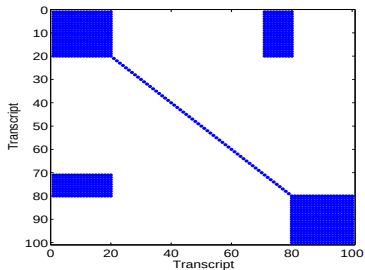
Simulation study 2

- We choose $p = 498$, $q = 100$ and $n = 120$.
- Consider 3 true predictors $\{30, 161, 239\}$. Associations between predictors and responses are generated according to following table:

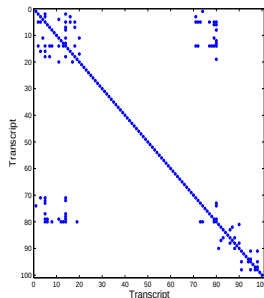
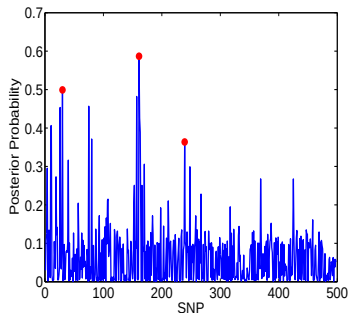
SNP ($\tilde{p}$)	Transcript ($\tilde{q}_p$)
30	1-20, 71-80
161	17-20
239	1-20, 71-80

- Corresponding elements of $\mathbf{B}$ have sd 0.3.
- Rest of the responses are simulated from noise with sd 0.1.

Simulation study 2: The true graph

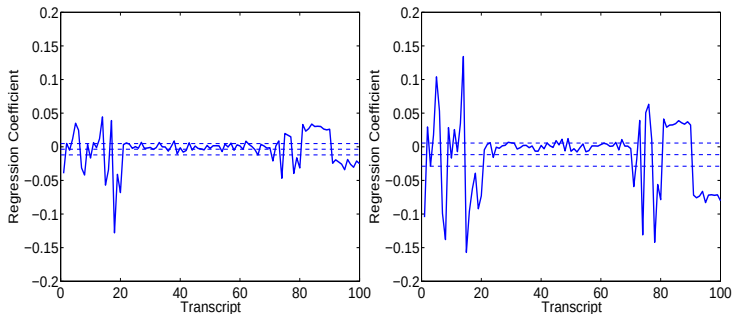


Results: Posterior probabilities



- Left: Posterior probabilities for γ , true variables circled in red.
- Right: Posterior probabilities for $\mathbf{G}$, with a cutoff on the posterior probabilities of edge inclusion set to 0.4

Results: Association analysis between SNPs and transcripts



- Left: Association of SNP 161 with all the 100 transcripts, showing enhanced association for transcripts 17-20.
- Right: association of SNP 239 with all the 100 transcripts, showing enhanced association for transcripts 1-20 and 71-80.

- Essentially, this is a regression problem where $\mathbf{X}$ = An $n \times p$ matrix of SNPs (Single Nucleotide Polymorphisms) and $\mathbf{Y}$ = An $n \times q$ matrix of gene expression data, for the same set of n individuals.
- An eQTL analysis tries to infer the $p \times q$ matrix $\mathbf{B}$, trying to associate genetic variability to the gene expressions.
- It's long been known that the genes are a part of a regulatory/interaction network.
- Statistically speaking, it is unreasonable to assume independence among the q traits.

Application to human eQTL analysis

- $n = 60$ unrelated individuals of Northern and Western European ancestry from Utah (CEU).
- SNP data publicly available from International Hapmap project (<http://hapmart.hapmap.org>).
- A total of $p = 3125$ SNPs found on 5' UTR of mRNA with minor allele frequency ≥ 0.1
- Gene expression data are also publicly available from the Sanger Institute website (<ftp://ftp.sanger.ac.uk/pub/genevar>).
- We work with $q = 100$ most variable transcripts out of a total of 47293.

- Controlling for FDR at 5% level yields 8 globally significant SNPs and 38 non-zero inverse covariance matrix elements.
- Yields a total of 43 significant associations.
- Chen et al. (2008, Bioinformatics) detected a slightly higher number of associations by considering both 3' and 5' UTRs simultaneously.
- Yields a total of 55 significant edges.

Open questions and current investigations

- Could the technique be extended to more flexible models, e.g. models that can handle a nonlinear mean function?
- Is it possible to show simultaneous variable and graph selection consistency?
- What about non-Bayesian approaches?

- ① **Bhadra, A.** and Mallick, B. K. (2012). Joint high-dimensional Bayesian variable and covariance selection with an application to eQTL analysis. (*under revision, Biometrics*)
- ② Dawid, A. P. and Lauritzen, S. L. (1993). Hyper Markov laws in the statistical analysis of decomposable graphical models. (*Ann. Statist. 21, 1272 - 1317*)
- ③ Lauritzen, S. L. (1996). Graphical Models. (*Oxford University Press*)